

EETS DevOps – Test 2 Preparation Guide

Printable study guide based on official test-prep outline.

1. Bash Scripting & Linux Shells

- Common shells: bash, sh, zsh, csh
- System configs: /etc/profile, /etc/shells
- User configs: ~/.bashrc, ~/.bash_profile, ~/.profile
- Check shell: echo \$SHELL
- Make script executable: chmod +x script.sh
- Debug scripts: bash -x script.sh
- Ctrl+C sends SIGINT
- Trap signals: trap 'command' SIGINT
- VAR=value (shell variable)
- export VAR=value (environment variable)

2. Ansible & Automation Fundamentals

- Agentless automation using SSH
- Inventory: list of managed hosts
- Playbook: YAML file of tasks
- Role: reusable structure
- Module: single unit of work
- Idempotency: same result on multiple runs
- Facts: system information
- Variable priority: task > host > play > global

Key Commands:

`ansible all -m ping`

`ansible-playbook site.yml`

`ansible-doc module_name`

`ansible localhost -m setup`

Jinja2 Templates:

`{{ variable }}` for dynamic configuration

3. YAML Syntax & Playbooks

- Indentation defines structure (spaces only)
- Lists use '-'
- Dictionaries use key: value
- Files often start with ---
- Execution is top-down
- vars_prompt used for runtime input

4. Cloud Computing & AWS Services

Cloud Concepts:

- Scalability, elasticity, cost efficiency
- IaaS provides infrastructure resources

Core AWS Services:

- EC2 – virtual servers
- S3 – object storage
- IAM – identity & access management
- VPC – networking
- SNS – notifications
- CloudFormation – infrastructure as code

S3 Storage Classes:

- Standard – frequent access
- IA – infrequent access
- Glacier Deep Archive – long-term storage

5. Exam Strategy & Tips

- Focus heavily on Ansible and AWS (highest weight)
- Expect scenario-based and conceptual questions
- Watch for YAML indentation errors
- Understand command outputs, not just commands
- Review CloudFormation vs Terraform differences

Tip: If you can explain a concept out loud, you are exam-ready.